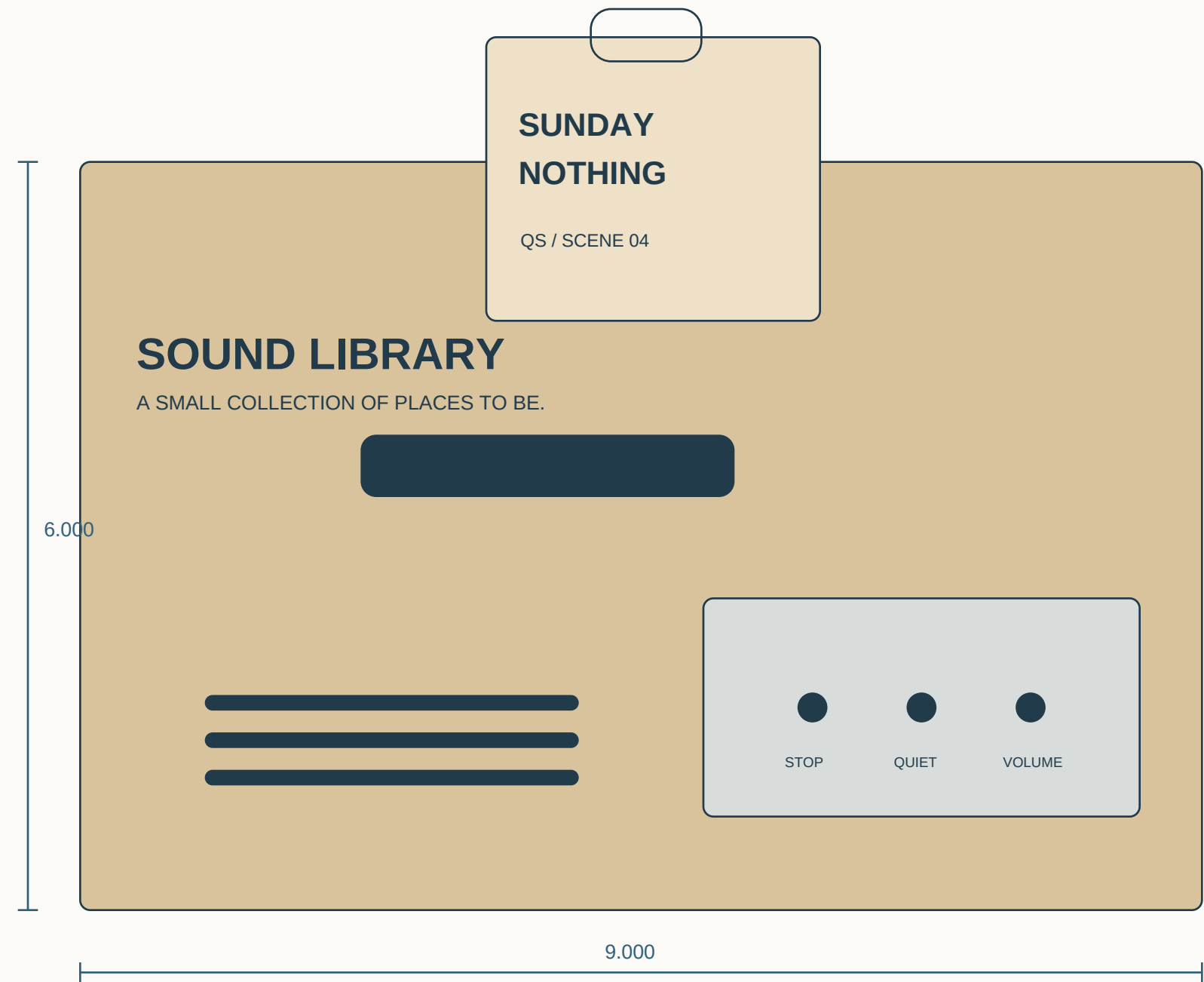




Body 9 W x 6 D x 3.75 H. Seated cartridge reaches Z5.375 before leather pull. This layout sketch is not a perspective measurement; use the orthographic sheets.

THE RITUAL

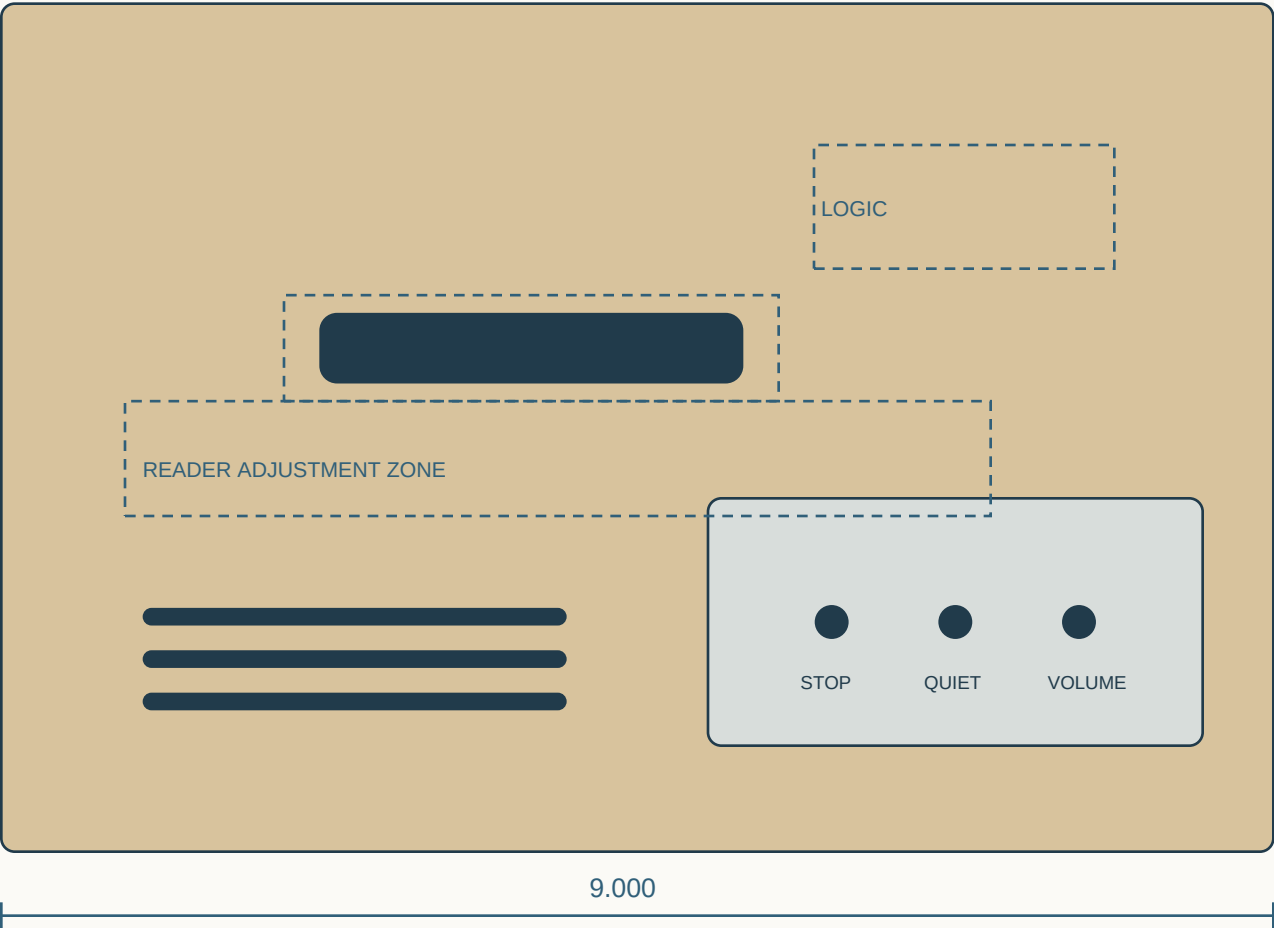
Insert a wooden cartridge. Its sound scene begins. Lift it out and the receiver stops. The cartridge identifies a local file; it contains no battery, music storage or account.

Four first scenes: Rain on Tin, Kitchen Disco, Last Train and Sunday Nothing. Humour lives in the titles and liner notes; the object itself can remain composed.

Solid walnut receiver; ash or walnut cartridges; small leather pull tabs; a modest metal control plate. A wooden thumb-grip works without leather.

Mono demonstration audio first. This package makes no hi-fi, gapless looping, commercial demand or retail-price claim.

This is the strongest system concept to test after the mood sign: new cartridges can add content without redesigning the receiver. That opportunity still needs real customer feedback.



PLAN: X left to right; Y front to rear. Blue zones are not pockets.

PART DATUMS

Lid 9 x 6 x 0.500. Slot X2.250 / Y3.3125; 3.000 x 0.500, corner R0.125. Cartridge 2.750 x 0.375 in section; loose trial fit, not a friction fit.

Four rings, each 9 x 6 x 0.750. Cavity X0.750 / Y0.750; 7.500 x 4.500, R0.125. Frame minimum wall 0.750. Grain along X throughout solid lid/rings.

Base: 9 x 6 x 0.250 actual plywood, surface mounted with eight slotted screw clearances. Body Z0 at underside; base top Z0.250, ring top/lid underside Z3.250, lid top Z3.750.

Grille: three slots X1.000; Y1.000, 1.300, 1.600; 3.000 x 0.125 with semicircular ends. Speaker behind lid in front-left bay, separate from reader.

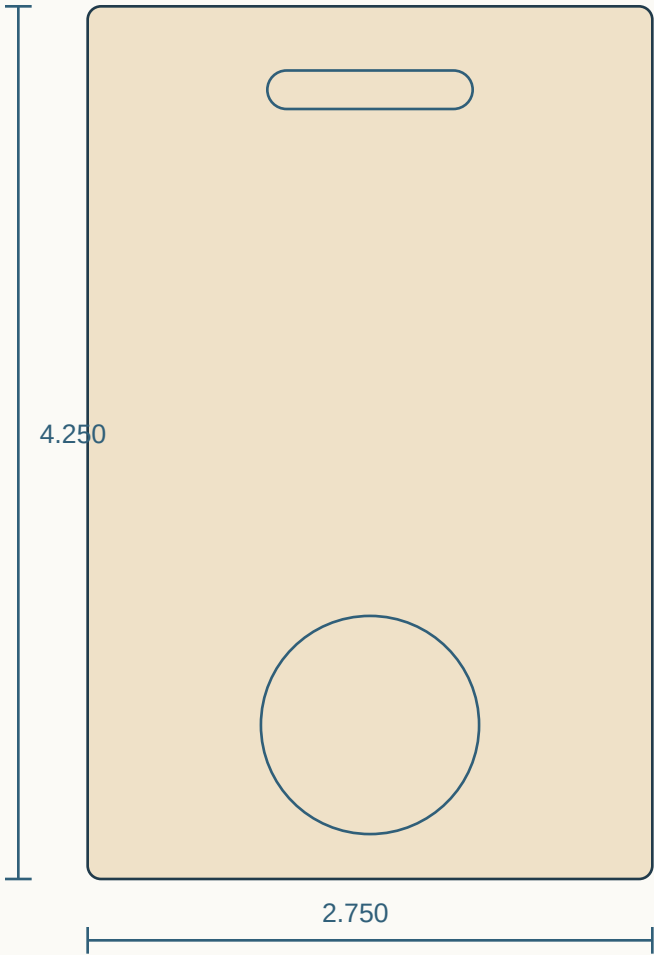
Control opening X5.250 / Y1.000; 3.000 x 1.250. Separate plate global X5.000 / Y0.750; 3.500 x 1.750. Actual STOP/toggle/pot bores held for samples.

SECTION / LAYER STACK



Initial 3-inch envelope failed the proposed vertical reader position. R01 increased body height to 3.750, leaving room to align the real antenna with the seated tag.

Glue grain-aligned rings and lid after the sensing jig passes. Removable base gives service access. Metal control plate, screws, controls and spare tags must be checked for RF interference in the finished assembly.



Cartridge thickness 0.375; outside R0.125.

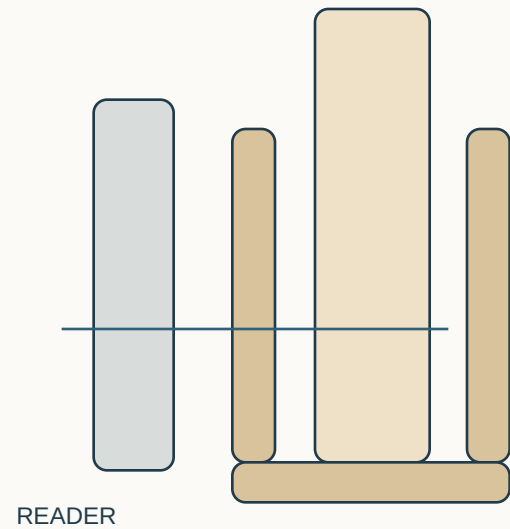
TAG FACE / TOWARD FRONT OF RECEIVER

Tag pocket centre: local U1.375 / V0.750, measured from cartridge bottom-left. Diameter 1.0625; depth 0.125. Leaves 0.250 wood. Selected tag: nominal 25 mm x 0.9 mm, sample-check before cutting.

Cover: wood disc 1.050 diameter x 0.0625, bonded at pocket mouth after RF and UID tests. Radial gap 0.00625 is a trial adhesive fit. Keep glue off tag and use nonmetallic cover. No screw or foil through antenna.

Pull slot: U0.875 / V3.750, 1.000 x 0.1875, end R0.09375. Leather <=0.750 wide and <=0.060 thick; loop and stitch, no metal rivet in the sensing region. Leather is optional.

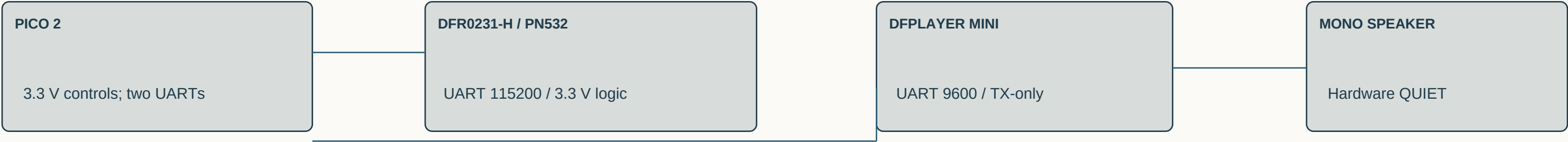
SIDE SECTION / GLOBAL Y-Z / SCHEMATIC



- Tag centre: X3.750 / Z1.875
- RF antenna target plane: Y3.160 (adjust to sample)
- Cartridge front face: Y3.375; base rests at Z1.125
- Guide front inner face Y3.3125; back Y3.8125

Guide: two 3.500 x 2.125 x 0.125 plywood faces; two 0.250 X x 0.500 Y x 2.125 Z end spacers; floor 3.500 x 0.750 x 0.250. Floor top Z1.125; guide top Z3.250. Outer X2.000..5.500 / Y3.1875..3.9375.

Optical beam crosses X through guide ends at global Y3.5625 / Z1.375. End-piece drawing uses Y horizontally and Z vertically; bore centre local 0.250 / 0.250, trial diameter 0.15625. Align actual emitter/receiver on adjustable brackets outside guide.



One fused 5 V supply feeds reader and audio; a diode branch feeds Pico VSYS. Use actual labels and wiring guide.

PICO PIN ASSIGNMENT

DFPlayer TX	GP0 / pin 1	1k series to player RX
NFC UART1 TX	GP4 / pin 6	reader C/R (RX)
NFC UART1 RX	GP5 / pin 7	reader D/T (TX)
Presence	GP6 / pin 9	optical OUT; 10k to 3V3
STOP / QUIET	GP7 / 10; GP8 / 11	10k pull-ups; see pole map
Volume	GP26 / pin 31	3V3 pot + ADC filter

Optical presence and RFID identity are separate. Accepted removal cancels queued sound. Unknown or changing UID stays silent. STOP/QUIET require fresh insertion; a reader timeout or framing fault requires power cycling with the slot empty.

Do not block controls behind a slow NFC library call. Non-blocking serial parsing, finite command timeouts and generation tokens are part of the firmware design. Pseudocode/model are not device firmware.

BUILD / CHECK SEQUENCE

1. Read four tag UUIDs on the bench.
2. Fit one cartridge + optical/reader jig.
3. Verify STOP/QUIET and removal during NFC reads.
4. Test audio and reader together for noise/resets.
5. Machine enclosure after the jig passes.
6. Repeat sensing tests inside finished enclosure.

One track per insertion, up to 180 seconds for the pilot. No streaming, gapless-loop or automatic-resume promise. Begin at low volume; the small speaker is for interaction testing.

Unplug external power and BOTH peripheral UART harnesses before USB programming. Never connect reader or player supply voltage to a Pico GPIO.

01 SOURCING + PHYSICAL FIT

Replaced the unavailable original reader; rejected contradictory sticker dimensions. Selected DFR0231-H and 25 mm disc tags. Increased case height after the vertical reader envelope failed the initial cavity. Replaced metal slot trim with a wood throat.

02 ELECTRICAL + CONTROL LOGIC

Separated cartridge presence from tag identity. Two UARTs prevent port collisions. Added transaction cancellation, fresh-insertion rules and fail-silent handling. Logic-model regression tests cover delayed reader results, removal, STOP, QUIET and faults.

03 PACKAGE + BUILDABILITY

Checked closed inch CAD geometry, part extents, capsule/slot corner fits, guide datums, lid/base stack and tag-cover depth. Reviewed rendered pages and assembly access. Added full-enclosure RF/audio test and explicit unverified purchase/fit items.

FIRST SHOP TASK: THE CARTRIDGE READER JIG

Use the 4.5 x 2.5 x 0.500 lid coupon, one cartridge and the plywood guide pieces. Position the reader and optical pair temporarily. Enrol the tag, test insertion/removal, then add nearby metal and the powered audio module. Record the successful antenna location before freezing mounts.

Pass target: 30 deliberate insertions/removals per cartridge, with no wrong selections or surprise restart; mixed-tag and bright-room trials; removal/STOP command issued within 100 ms in measured firmware. These are future bench acceptance targets, not completed tests.

Three internal reviews improve the draft; they do not substitute for physical fit, RF range, electrical compliance, firmware-on-hardware tests or a Mastercam import. Current package is R03; earlier decision snapshots are retained as superseded history.